AC4409

Corporate Financing

Winter 2022 Examination
Paper 1 · 1.5 hours · Answer 2 of 3 questions · Show ALL workings

Module	AC4409 — Corporate Financing
Session	Winter 2022
Structure	Three questions, each worth 50 marks. Answer any two — all questions carry equal weight.
This scheme covers	All three questions in full — Q1, Q2, and Q3 — so you can study every possible pairing, not just the two you'd pick on the day.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the method marks for those later steps. This applies throughout the reorder-level and cost-of-capital workings below.

A source-paper note. Question 3(d)'s CianPa plc balance sheet extract doesn't quite reconcile — total assets less total liabilities comes to €61,037m by the figures given, not the stated €51,524m net assets. This looks like a transcription artefact in the source table rather than something to solve for; since gearing ratios only need the borrowings, net assets, EBIT, interest and market capitalisation figures directly (not a fully reconciled balance sheet), the ratios below are calculated straight from those figures regardless.

Question 1

Cash Conversion Cycle, Credit Policy & Inventory Reorder Levels (50 Marks)

Part (a)(i)

5 Marks

(i) What is the Cash Conversion Cycle (CCC) and how does it relate to working capital management?

FULL MODEL ANSWER

The Cash Conversion Cycle measures the length of time, in days, that cash is tied up in a company's operating cycle before it's converted back into cash from customers.

CCC = Days Inventory Outstanding (DIO) + Days Sales Outstanding (DSO) − Days Payables Outstanding (DPO)

Relationship to working capital management. The CCC is the single most direct, trackable measure of how efficiently a company is managing its working capital. A shorter (or more negative) CCC means less cash is tied up in the gap between paying suppliers and collecting from customers, freeing that cash for other uses; a longer CCC means the company needs more external financing (overdraft, short-term loans) just to fund its day-to-day operating cycle. Because the CCC decomposes into its three separate drivers (inventory, receivables, payables), it also works as a diagnostic tool: a rising CCC over time signals exactly where working capital is deteriorating — whether stock is moving more slowly, customers are taking longer to pay, or the company itself is paying suppliers faster than it needs to — letting management target the specific lever that needs attention.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Correctly states the CCC formula with all three components, and explains its role as both a measure of working capital tied up and a diagnostic tool for targeting specific improvements.
2–3	2–3	Correct formula but limited explanation of its practical role in working capital management.
0–1	0–1	No attempt, or an incorrect or incomplete formula.

Part (a)(ii) **5 Marks**

(ii) Outline THREE factors that a company should consider before giving credit to a new customer.

FULL MODEL ANSWER

Credit assessment is commonly organised around the “5 Cs of credit” — three of the most important for a new customer are:

1. Character — the customer's payment history and reputation. Checking trade references from the customer's other suppliers, and reviewing any available credit history or payment record, gives a sense of whether the customer has a track record of paying on time or chronically delaying payment.

2. Capacity — the customer's financial ability to pay. Reviewing the customer's financial statements (where available), cash flow position, and existing debt obligations helps assess whether the customer's business genuinely generates enough cash to meet the proposed credit terms, rather than simply assuming good intentions will translate into actual payment.

3. Conditions — the broader economic and industry environment the customer operates in. A customer trading in a currently struggling sector, or in a period of tightening economic conditions generally, carries a higher risk of payment difficulty even if their own historical record looks reasonable — external conditions can deteriorate a previously sound credit risk quickly.

Other valid answers from the same framework include **capital** (the customer's own financial resources/net worth as a buffer) and **collateral** (security available if the customer defaults) — either would also be creditable as one of the three factors, provided it's clearly explained.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Identifies three genuinely distinct factors (from the 5 Cs framework or equivalent) with a clear explanation of what each involves assessing.
2–3	2–3	Two factors clearly explained, or three named with limited explanation.
0–1	0–1	No attempt, or factors named with no explanation at all.

Part (b) **10 Marks**

Calculate the Cash Conversion Cycle for Tara Ltd using the data in Table 1.

	Tara Ltd (€)
Revenue	850,000
Cost of sales	500,000
Current assets	
Inventory	100,000
Trade receivables	84,000
Current liabilities	
Trade payables	107,000

FULL MODEL ANSWER

Component	Formula	Calculation	Days
DIO	Inventory ÷ Cost of sales × 365	100,000 ÷ 500,000 × 365	73.00
DSO	Receivables ÷ Revenue × 365	84,000 ÷ 850,000 × 365	36.07
DPO	Payables ÷ Cost of sales × 365	107,000 ÷ 500,000 × 365	78.11

CCC = DIO + DSO − DPO
= 73.00 + 36.07 − 78.11
≈ 30.96 days

DSO uses **Revenue** as its base (receivables represent amounts owed on sales made at selling price), while DIO and DPO both use **Cost of sales** as their base (inventory and payables both relate to goods at their cost to Tara, not their eventual selling price) — mixing these bases up is the most common error in this calculation. Tara's CCC of ≈31 days means it takes about a month, on average, from paying for inventory to receiving cash from the eventual sale, after allowing for the credit period it receives from its own suppliers.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly calculates DIO, DSO (using revenue, not cost of sales), and DPO, and correctly combines them to reach ≈30.96 days.
5–7	5–7	Correct method but one component uses the wrong base (e.g. cost of sales instead of revenue for DSO), carried through under the own-figure rule.
2–4	2–4	Correctly calculates one or two of the three components but doesn't combine them correctly into the CCC.
0–1	0–1	No attempt, or no understanding of the CCC formula.

Loona Ltd has an Economic Order Quantity (EOQ) of 536 units and reorders every 8 days. Its usage is constant per day. The supplier typically takes five days to deliver the goods.

Part (c)(i) — 5 Marks

(i) You are informed that the lead time on purchases may now vary between four and six days. Calculate the reorder level which should ensure that there are no stock outs.

FULL MODEL ANSWER

Daily usage = EOQ ÷ reorder cycle = 536 ÷ 8
= 67 units per day

Reorder level = Maximum lead time × daily usage
= 6 days × 67 units/day
= 402 units

Using the **maximum** lead time (6 days) guarantees stock lasts through the worst-case delivery delay; using the average (5 days) would leave the business exposed to a stock-out whenever delivery happens to fall on the slower end of the range.

Part (c)(ii) — 5 Marks

(ii) You are informed that sales demand for the inventory can range from 306 to 752 units every 8 days. Calculate the reorder level which should ensure that there are no stock outs.

FULL MODEL ANSWER

Maximum daily usage = 752 ÷ 8 = 94 units/day
(Lead time reverts to the original, constant 5 days for this part)

Reorder level = Lead time × MAXIMUM daily usage
= 5 days × 94 units/day
= 470 units

Here it's daily usage, not lead time, that's uncertain — so the reorder level needs to cover the **maximum** possible usage (94 units/day) over the known, constant 5-day lead time, to guard against the worst case of unusually high demand arriving during a normal delivery wait.

Part (c)(iii)

5 Marks

(iii) Assume that both uncertainties outlined in (i) and (ii) exist. Calculate the reorder level which should ensure that there are no stock outs.

FULL MODEL ANSWER

Reorder level = MAXIMUM lead time × MAXIMUM daily usage
= 6 days × 94 units/day
= 564 units

When both lead time and usage are uncertain simultaneously, guaranteeing no stock-outs requires covering the single worst combination of both — the longest possible delay (6 days) *and* the highest possible usage rate (94 units/day) occurring together. Using the maximum of only one variable while leaving the other at its average or base level would still leave a residual stock-out risk in the scenario where both happen to turn unfavourable at once.

How the 15 Marks for Part (c) Are Likely Allocated

Tier	Marks	What separates this tier
(i) 5	5	Correctly calculates daily usage (67 units/day) and applies the maximum lead time (6 days) to reach 402 units.
(ii) 5	5	Correctly calculates maximum daily usage (94 units/day) and applies the constant 5-day lead time to reach 470 units.
(iii) 5	5	Correctly combines the maximum lead time AND maximum daily usage from parts (i) and (ii) to reach 564 units, rather than only varying one of the two factors.

Part (d) **15 Marks**

Enqvist, Graham and Nikkkinen (2014) find that “the impact of the business cycle on the working capital–profitability relationship is more pronounced in economic downturns relative to economic booms.” How did they reach this conclusion? And what is the implication of this finding for working capital management?

Reference: Enqvist, Julius, Michael Graham, and Jussi Nikkinen. “The impact of working capital management on firm profitability in different business cycles: Evidence from Finland.” *Research in International Business and Finance* 32 (2014): 36–49.

FULL MODEL ANSWER

How they reached this conclusion

The data and setting. The authors used a panel dataset of Finnish listed companies spanning an 18-year period — a sufficiently long window to capture multiple full business cycles, including the 2007–2008 downturn that motivated the study.

The working capital measure. Working capital efficiency was measured using the Cash Conversion Cycle (CCC) and its individual components (days inventory outstanding, days sales outstanding, and days payables outstanding) — the same decomposition used elsewhere in this paper (see part (a)(i)).

The empirical method. They ran panel data regressions of firm profitability (return on assets) on the CCC and its components, following an approach similar to earlier working-capital-profitability studies (such as Deloof, 2003, and Lazaridis and Tryfonidis, 2006), but split the analysis by economic conditions — comparing the estimated relationship between working capital and profitability separately for economic downturn periods versus economic upturn (boom) periods within their sample, while controlling for other firm characteristics known to affect profitability.

The comparison that produced the finding. Comparing the size and statistical significance of the CCC–profitability relationship across the two regimes, they found a negative relationship overall (shorter CCC associated with higher profitability, consistent with most of the prior literature), but with a **materially larger and more significant negative coefficient during economic downturns** than during booms — specifically, they showed that the significance of efficient inventory management and accounts receivable collection increases during periods of economic downturn.

The implication for working capital management

Working capital discipline pays off disproportionately in bad times. During a downturn, external financing typically becomes scarcer and more expensive as credit conditions tighten — banks lend more cautiously and at higher spreads, and equity markets are less receptive. In that environment, cash freed up through efficient working capital management (faster inventory turnover, quicker receivables collection) effectively becomes an internal substitute for increasingly expensive or unavailable external financing, which is exactly why its impact on profitability is larger when times are hard.

Working capital policy shouldn't be treated as static across the cycle. The practical implication for a financial manager or CFO is that working capital management deserves particular attention and discipline heading into, and during, an economic downturn — tightening credit terms, accelerating collections, and managing inventory more actively isn't just good practice generally, it's specifically where the biggest profitability payoff is available precisely when the business can least afford to be complacent about it. In a boom, when credit is abundant, the same degree of working capital discipline still helps, but the marginal benefit to profitability is smaller.

A strong answer explicitly connects the empirical method (panel regression, split by business cycle) to the specific finding (larger, more significant negative coefficient in downturns), rather than only restating the conclusion in different words — the question asks *how* they reached the conclusion, not just what the conclusion was.

How the 15 Marks Are Likely Allocated

Tier	Marks	What separates this tier
12–15	12–15	Correctly describes the panel data methodology (Finnish listed firms, 18-year period, CCC-based measure, comparison across business cycle phases), AND draws a clear, well-reasoned implication connecting scarce/expensive external financing in downturns to the greater value of internally-generated working capital efficiency.
8–11	8–11	Correctly restates the finding and gives a reasonable implication, but with limited detail on how the authors actually reached the conclusion (i.e. the methodology itself).
4–7	4–7	General discussion of working capital and business cycles without specific engagement with this paper's actual approach or finding.
0–3	0–3	No attempt, or a answer unconnected to the cited paper.

Question 1 Total: 50 Marks

Question 2

Factoring & Venture Finance (50 Marks)

Susie Plc has been set up for the purpose of importing commodities that will be sold to a small number of reliable customers. Sales invoicing is forecast at €300,000 per month. The average credit period for this type of business is 2.5 months.

The company is considering factoring its accounts receivable under a full factoring agreement without recourse (i.e., non-recourse factoring). Under this agreement, the factor will charge a fee of 2.5% on total invoicing. He will give an advance of 85% of invoiced amounts, at an interest rate of 13% per annum.

The agreement should enable Susie Plc to avoid spending €95,000 per annum on administration costs.

Part (a)(i)

5 Marks

(i) Calculate the annual net cost of factoring.

FULL MODEL ANSWER

Step 1 — The scale of financing involved

Annual invoicing = €300,000 × 12 months = €3,600,000
Average trade receivables = €300,000 × 2.5 months = €750,000
Amount of finance provided by factor (85% advance) = 85% × €750,000 = €637,500

Step 2 — Costs and savings

	€
Factor's fee (2.5% × €3,600,000 annual invoicing)	90,000
Factor's interest on advance (13% × €637,500)	82,875
Less: administration cost savings	(95,000)
Net annual cost of factoring	**77,875**

The factor's fee is charged on **total annual invoicing** (€3,600,000), but the interest charge only applies to the **85% advance actually drawn** (€637,500) — these two percentages are applied to two genuinely different bases, and mixing them up is the most common error in this type of question.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Correctly calculates average receivables and the 85% advance amount, correctly applies the fee to total invoicing and the interest rate to the advance only, and nets off the admin savings to reach €77,875.
2–3	2–3	Correct method but the fee and interest rate applied to the wrong (same) base, or the admin savings omitted.
0–1	0–1	No attempt, or no structured cost calculation.

Part (a)(ii) **5 Marks**

(ii) Discuss the financial benefits of such an agreement, having regard to a current interest rate on bank overdrafts of 6%. Assume 85% of the invoiced amount is the amount of finance required.

FULL MODEL ANSWER

Isolating just the financing element (the €637,500 advance) and comparing the factor's rate against the alternative of financing the same amount through a bank overdraft:

	€
Cost of financing €637,500 via the factor's advance (13%)	82,875
Cost of financing the SAME €637,500 via a bank overdraft (6%)	38,250
Extra cost of the factor's financing over a plain overdraft	**44,625**

The factor's financing is, in isolation, considerably more expensive. At 13%, more than double the 6% overdraft rate, the pure financing cost of the factor's advance is €44,625 a year more expensive than simply borrowing the same amount via overdraft. On the financing dimension alone, factoring looks like a poor deal.

But the fee and admin savings roughly offset each other. The factor's 2.5% fee (€90,000) is actually slightly less than the €95,000 administration cost it replaces — a small net saving of €5,000 on that dimension, before considering financing at all.

Taken together, the overall arrangement costs more than the alternative. Combining both effects, the net annual cost of €77,875 (from part (i)) is substantially worse than simply keeping administration in-house and financing receivables via a 6% overdraft, which would cost roughly €38,250 in financing with the €95,000 admin cost continuing to be incurred either way. The genuine benefit of non-recourse factoring isn't really the financing rate at all — it's the transfer of bad debt risk to the factor (since it's *without recourse*) and the freeing up of management time from credit control, which aren't fully captured in this purely financial comparison.

A complete answer notes that the raw numbers argue against the deal on financing cost grounds, but a full recommendation would also weigh the non-recourse bad debt protection and management time savings, which don't show up as a euro figure in this calculation but may still be valuable to Susie Plc given it's a new company built around a small number of customers — concentration risk that non-recourse factoring genuinely does mitigate.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Correctly compares the 13% factor financing cost against the 6% overdraft alternative on the same €637,500 base, quantifies the €44,625 difference, and discusses the qualitative benefits (non-recourse risk transfer) not captured in the pure cost comparison.
2–3	2–3	Correctly identifies that the factor's rate is more expensive than the overdraft but without quantifying the difference or discussing the qualitative trade-offs.
0–1	0–1	No attempt, or no comparison made to the 6% overdraft rate at all.

Part (a)(iii) **5 Marks**

(iii) Critically evaluate ONE other form of short-term financing available to Susie Plc.

FULL MODEL ANSWER

Bank overdraft is a natural comparator, given it's already referenced in part (ii).

> **Advantages for Susie Plc specifically.** An overdraft is far cheaper (6% vs the factor's effective cost) and gives Susie complete control over its own credit control and customer relationships, which may matter given the business is built around a small number of reliable, presumably relationship-based customers — handing collections to a third-party factor could be seen as an unwelcome intrusion on those relationships.

> **Disadvantages for Susie Plc specifically.** An overdraft provides no protection against bad debts (unlike non-recourse factoring), leaving Susie fully exposed if one of its small number of customers defaults — and with a concentrated customer base, that risk is more material than it would be for a business with many small customers. An overdraft is also technically repayable on demand, and as a newly set-up company Susie may find banks reluctant to extend a substantial facility without a trading track record, which factoring (secured against the receivables themselves rather than Susie's own creditworthiness) doesn't require to the same degree.

Trade credit from Susie's own suppliers of the commodities it imports is another reasonable alternative to evaluate, particularly relevant given Susie's business model is built on importing — but whichever alternative is chosen, a strong answer weighs it specifically against Susie's situation (new company, concentrated customer base, import business) rather than describing the financing tool generically.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Critically evaluates one genuinely short-term financing alternative with advantages and disadvantages specifically tied to Susie Plc's situation (new company, small customer base, import business).
2–3	2–3	A relevant alternative identified with generic advantages/disadvantages not tied to Susie's specific circumstances.
0–1	0–1	No attempt, or a source of financing that isn't genuinely short-term.

Part (b)(i) **5 Marks**

(i) Using the EnviroPump case as an example, what are the challenges of self-financing a start-up?

FULL MODEL ANSWER

This question refers to a specific case study (EnviroPump) covered in your lectures, whose full facts aren't reproduced on this exam paper. The model answer below covers the general challenges of self-financing (bootstrapping) a start-up that any well-prepared answer should include; you should slot in the specific EnviroPump details from your course materials (founders' personal financial position, timeline, specific funding gaps mentioned in the case) as the examples the question is actually asking for.

Limited personal capital caps the pace of growth. A founder self-financing (bootstrapping) a start-up is constrained by their own savings and the cash the business itself generates — there's no external capital injection to accelerate hiring, marketing, or production capacity beyond what current revenue can support, which can mean losing ground to better-funded competitors or missing a window of market opportunity.

Reinvesting profits limits the founder's own financial cushion. Every euro ploughed back into the business is a euro the founder isn't taking out for their own living costs or as a personal safety buffer, which can create real personal financial strain, especially in the early, cash-hungry years before the business reaches self-sustaining profitability.

No access to an investor's expertise, network, or credibility signal. Beyond just capital, external investors (particularly venture capitalists or experienced angels) typically bring industry contacts, strategic guidance, and a credibility signal to future customers, partners, and later-round investors. A self-financed start-up forgoes all of this, relying entirely on the founder's own network and judgement.

Concentrated personal risk. Self-financing typically means the founder's personal wealth (savings, sometimes property used as collateral for a personal loan) is directly at risk if the business fails, rather than that risk being shared with outside investors who've knowingly taken on venture-style risk in exchange for equity.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Identifies multiple genuine challenges of self-financing (growth pace constraints, personal financial strain, lack of external expertise/network, concentrated personal risk) with specific reference to the EnviroPump case facts covered in lectures.
2–3	2–3	General challenges of self-financing identified correctly but without reference to the specific EnviroPump case.
0–1	0–1	No attempt, or no genuine engagement with self-financing challenges.

Part (b)(ii) **7 Marks**

(ii) What is a term sheet and how does it help entrepreneurs to choose between different potential investors?

FULL MODEL ANSWER

A term sheet is a short, non-binding document setting out the proposed key terms of an investment before the full legal agreements are drafted — effectively a summary of the deal both sides intend to move forward with.

> **What a term sheet typically covers.** Valuation (pre-money and post-money), the amount being invested and the resulting ownership percentage, the type of security being issued (ordinary shares, preferred shares, convertible notes), liquidation preference (who gets paid first, and how much, in an exit or wind-down), anti-dilution protection, board composition and voting/veto rights, and often founder vesting schedules and the size of any option pool reserved for future employees.

> **Why it helps entrepreneurs compare investors, beyond just the valuation number.** A headline valuation figure alone is a poor basis for comparing offers, because two investors offering the “same” valuation can propose very different actual terms underneath it — one demanding a substantial liquidation preference (which can leave founders and common shareholders with far less than the valuation implies in a modest exit) or extensive board control and veto rights, while another offers cleaner, founder-friendly terms at the same headline number. The term sheet exposes these differences early, before significant legal costs are sunk into a full agreement, letting an entrepreneur properly compare the economics and control implications of competing offers side by side.

> **It also serves as a negotiating and signalling tool.** Because it's typically non-binding on the commercial terms (though certain clauses like exclusivity and confidentiality usually are binding), a term sheet lets both sides test whether they can agree on the big-picture deal points before committing to the time and legal expense of full documentation, and receiving a term sheet from a credible investor can itself help an entrepreneur negotiate better terms with other prospective investors.

How the 7 Marks Are Likely Allocated

Tier	Marks	What separates this tier
6–7	6–7	Correctly defines a term sheet, lists several of its key components (valuation, liquidation preference, board rights, anti-dilution), and explains specifically why comparing full term sheets (not just headline valuation) matters for choosing between investors.
3–5	3–5	Correct general definition but with limited detail on specific components, or limited explanation of why it helps compare investors specifically.
0–2	0–2	No attempt, or confuses a term sheet with a fully binding legal agreement.

Part (b)(iii) **8 Marks**

(iii) What financial advice would you give to a young, small firm which has a high burn rate, a short runway and an exciting high-sought after recently newly patented product?

FULL MODEL ANSWER

The immediate priority is extending the runway, not necessarily growing faster. With a short runway (limited months of cash left at the current burn rate), the firm's most urgent task is buying itself more time before it runs out of cash entirely — even a genuinely exciting, in-demand patented product doesn't help if the company can't survive long enough to capitalise on it.

Cut the burn rate directly wherever possible, without damaging the core asset. Reviewing discretionary spending, delaying non-essential hires, and focusing resources specifically on whatever activity most directly protects or monetises the patent (rather than broader growth initiatives) extends the runway without needing to raise a euro of new capital.

Consider a bridge financing instrument rather than a full priced equity round under time pressure. A convertible note or SAFE (simple agreement for future equity) can raise cash quickly without forcing an immediate, difficult valuation negotiation while the company is in a weak bargaining position due to its short runway — the valuation gets resolved later (typically at the next full funding round), which avoids locking in a low valuation purely because the firm is under cash pressure right now.

The patent itself is a genuine, leverageable asset — use it as one. A newly patented, high-demand product gives the firm real options beyond a traditional equity raise: licensing the technology to a larger, better-capitalised partner in exchange for upfront licensing fees or royalties, or using the patent as collateral for asset-backed lending, can generate cash without further diluting the founders, and a licensing deal can also validate the technology commercially in a way that strengthens the firm's position in any subsequent equity negotiation.

Avoid a distressed "down round" if at all possible. Raising equity from a position of obvious weakness (a near-empty runway) typically means accepting a much lower valuation and harsher terms (larger liquidation preferences, more investor control) than the underlying quality of the patented product would otherwise command — which is exactly why buying time first (via cost cuts or bridge financing) before approaching equity investors properly usually produces a better outcome than raising immediately under duress.

Explore non-dilutive funding sources specific to innovative, patented products. Grants, R&D tax credits, and innovation-focused government schemes are worth pursuing specifically because they don't require giving up any equity at all, and a firm with a genuinely patented, high-demand product is often a strong candidate for this kind of support.

How the 8 Marks Are Likely Allocated

Tier	Marks	What separates this tier
6–8	6–8	Identifies the urgency of extending the runway, recommends a specific appropriate financing instrument for the situation (bridge note/SAFE rather than a full priced round under pressure), and explicitly leverages the patent as an asset (licensing, IP-backed financing) rather than treating it as incidental background detail.
3–5	3–5	Sensible general advice (cut costs, raise money) without specifically addressing why a bridge instrument suits the short-runway situation or how the patent itself can be used strategically.
0–2	0–2	No attempt, or generic advice with no connection to the specific high-burn/short-runway/patented-product scenario.

Part (b)(iv)

15 Marks

(iv) Seven years after receiving venture finance, EnviroPump has grown in size (4,000+ employees) and value (yearly profits of $3 million). The company is considering listing on the stock market.

The firm is considering listing on the AMEX. Critically assess the arguments for and against listing on the stock market.

FULL MODEL ANSWER

Arguments FOR Listing

Access to a much larger pool of capital for future growth. A public listing opens access to public equity markets far beyond what private venture or growth-equity investors alone could provide, supporting further expansion, acquisitions, or R&D investment without relying on repeated private funding rounds.

Liquidity and an exit route for existing venture investors and early employees. Seven years after the original venture financing, the original investors are likely looking for a route to realise their return — a listing provides exactly that liquidity, and stock options for the 4,000+ employees become a tangible, tradeable benefit rather than a purely notional one.

Enhanced credibility and public profile. Being publicly listed can improve standing with customers, suppliers, and future business partners, and gives the company a public equity currency (its own tradeable shares) that can be used to fund future acquisitions.

Arguments AGAINST Listing

Substantial listing and ongoing compliance costs. The upfront costs of a listing (underwriting, legal, accounting, and exchange fees) are significant, and ongoing public company obligations (financial reporting, disclosure requirements, governance standards) impose a permanent additional cost and management time burden that a private company doesn't carry.

Loss of control and short-term market pressure. Public shareholders and analysts often push for short-term performance (quarterly results focus) which can pressure management away from longer-term strategic investments, and the founders' and early investors' control is diluted further and subject to public market scrutiny and potential activist shareholder pressure in a way private ownership isn't.

Whether AMEX specifically is the right venue for a company of this profile is a genuine question. AMEX (now NYSE American) has historically served smaller-cap and growth companies, so it may suit EnviroPump's current size reasonably well — but $3 million in yearly profit against 4,000+ employees implies fairly thin per-employee profitability, which is worth scrutinising before a listing: public markets and analysts will likely probe margins and profitability trends closely, and a company with relatively thin current profitability relative to its headcount may face a tougher reception, or a lower valuation, than the growth story alone would suggest. This is a case where being ready in principle (size, employee count) doesn't automatically mean being ready in practice (profitability profile, governance readiness for public company obligations).

A strong answer doesn't treat this as a generic "listing pros and cons" essay — it engages with EnviroPump's specific numbers (4,000+ employees against only $3m profit is a fairly thin margin profile for a company of that scale) and with the specific choice of exchange (AMEX/NYSE American, historically a smaller-cap venue), reaching a reasoned view on readiness rather than just listing generic arguments on both sides.

How the 15 Marks Are Likely Allocated

Tier	Marks	What separates this tier
12–15	12–15	Covers multiple genuine arguments on both sides (capital access, investor liquidity, and credibility for; cost/compliance burden, loss of control, and short-term pressure against), AND critically engages with EnviroPump's specific profile (the thin profit-to-headcount ratio, and whether AMEX specifically suits a company of this size) rather than giving a generic answer.
8–11	8–11	Covers several arguments on both sides adequately but without engaging with EnviroPump's specific numbers or the AMEX-specific consideration.
4–7	4–7	One-sided discussion (only arguments for, or only against), or very generic treatment of listing pros and cons.
0–3	0–3	No attempt, or no genuine critical assessment.

Question 2 Total: 50 Marks

Question 3

Capital Structure Theory, Cost of Capital & Gearing (50 Marks)

Part (a) **20 Marks**

Is the trade-off model the best model of capital structure? Why?

FULL MODEL ANSWER

No single capital structure theory deserves the title of definitively "best" — the trade-off model captures something real and important, but it has genuine competitors that explain patterns it struggles with.

Partially, but not conclusively better than its main rivals

The case FOR the trade-off model

It gives a clean, intuitive economic logic for why an interior optimal leverage exists. By balancing the tax benefit of debt against the expected costs of financial distress, the trade-off model produces a sensible prediction: firms should have SOME debt (to capture the tax shield) but not unlimited debt (because distress costs eventually dominate) — and it gives a clear economic mechanism for why the optimal point sits where it does.

It's broadly consistent with cross-industry leverage patterns. Industries with low distress costs and stable cash flows (regulated utilities, real estate) do tend to carry more debt than industries with high distress costs and volatile cash flows (technology, pharmaceuticals) — exactly what the trade-off model would predict.

Where the trade-off model struggles

It doesn't explain why the most profitable firms often carry the LEAST debt. The trade-off model predicts more profitable firms (which face lower expected distress costs, and can make fuller use of the tax shield since they have taxable income to shield) should carry MORE debt. In practice, highly profitable firms are frequently observed borrowing very little, financing growth internally instead — the opposite of what a pure trade-off logic predicts. The pecking order theory (Myers, 1984) explains this pattern far better: profitable firms simply have more internal funds available, so they need external financing (of any kind) less often, regardless of what an "optimal" leverage target might say.

It implies firms actively target a specific leverage ratio, which the evidence doesn't clearly support. The trade-off model's core prediction is that firms adjust toward an optimal target ratio over time. Empirically, firms' capital structures often look more like the cumulative, path-dependent result of past financing decisions (issue debt when internal funds run short, issue equity only as a last resort) than the result of actively rebalancing toward a fixed target — again, more consistent with pecking order behaviour.

It says little about market timing effects. Firms appear to time their financing choices based on market conditions — issuing equity disproportionately when share prices are high (market timing theory) — a pattern the trade-off model, focused purely on tax shields and distress costs, has no natural mechanism to explain.

A reasoned conclusion

The trade-off model is best understood as one lens among several, not a complete theory on its own. It correctly identifies real forces (the tax shield, distress costs) that genuinely matter and that show up clearly in cross-industry leverage patterns, but it's demonstrably weaker than the pecking order theory at explaining firm-level financing behaviour over time, particularly the profitability-leverage relationship. A complete answer to "what determines capital structure" likely needs elements of both — and of agency cost theory and market timing too — rather than treating any single model as sufficient on its own.

A candidate arguing the trade-off model IS the best model could reasonably do so by emphasising its clear economic logic and its success explaining cross-industry (rather than cross-firm) leverage patterns — either position is defensible provided it engages seriously with the model's actual empirical strengths and weaknesses rather than simply asserting a view.

How the 20 Marks Are Likely Allocated

Tier	Marks	What separates this tier
15–20	15–20	Takes a clear position, explains the trade-off model's genuine strengths (cross-industry leverage patterns, tax shield vs. distress cost logic), AND engages seriously with its key empirical weakness (the profitability-leverage relationship, better explained by pecking order theory), reaching a reasoned overall conclusion.
10–14	10–14	Correctly explains the trade-off model's mechanics and cites at least one competing theory, but without a clear, well-supported position on whether it's genuinely "best".
5–9	5–9	General description of the trade-off model without meaningful comparison to alternative capital structure theories.
0–4	0–4	No attempt, or a fundamental misunderstanding of what the trade-off model actually claims.

Part (b) — Scenario

DebPro plc is trying to introduce an improved method of assessing investment projects using discounted cash flow techniques. For this it must obtain a cost of capital to use as a discount rate. The finance department has assembled the following information:

- The company has an equity beta of 1.50. The yield on risk-free government securities is 7% and the historic premium above the risk-free rate is 5% for shares.
- The market value of the firm's equity is twice the value of its debt.
- The cost of borrowed money to the company is estimated at 12% (before tax shield benefits). Corporation tax is 30%.

Assume: No inflation. Best guess IRR_1 = 8%.

Part (b)(i)

5 Marks

(i) Estimate the equity cost of capital using the capital asset pricing model (CAPM).

FULL MODEL ANSWER

$K_e = R_f + \beta(\text{Risk premium})$
= 7% + 1.50 × 5%
= 7% + 7.5%
K_e = 14.5%

The "Best guess IRR_1 = 8%" and the annuity table/formula sheet supplied with this paper relate to a DCF/IRR-interpolation calculation that isn't actually required by parts (i) or (ii) as set — they appear to be context or formula reference material for a further part of this question not reproduced here (or a leftover from a related question bank). Neither the CAPM calculation above nor the WACC calculation in part (ii) needs the 8% IRR figure or the annuity table at all.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Correctly applies CAPM with the given inputs to reach 14.5%.
2–3	2–3	Correct formula but an arithmetic slip, or confuses the 5% premium with a market return rather than a premium above the risk-free rate.
0–1	0–1	No attempt, or the wrong formula used.

Part (b)(ii) **5 Marks**

(ii) Create an estimate of the weighted average cost of capital (WACC).

FULL MODEL ANSWER

Equity is twice the value of debt, so: E = 2 parts, D = 1 part, Total = 3 parts
Weight of equity = 2/3 = 66.67%   Weight of debt = 1/3 = 33.33%

WACC = (E/V) × K_e + (D/V) × K_d × (1−T)
= (2/3 × 14.5%) + (1/3 × 12% × (1−30%))
= 9.67% + (1/3 × 8.4%)
= 9.67% + 2.80%
WACC ≈ 12.47%

"Equity is twice the value of debt" means E = 2D, not E = 2 × (D+E) or any other combination — setting up the weights as E/(E+D) = 2/3 and D/(E+D) = 1/3 is the key step. As in Q3(b) of other AC4409 papers, the 12% cost of debt is pre-tax and must be adjusted for the tax shield before being weighted into WACC.

How the 5 Marks Are Likely Allocated

Tier	Marks	What separates this tier
4–5	4–5	Correctly derives the 2/3 : 1/3 equity/debt weighting from "equity is twice debt", applies the tax shield to the cost of debt only, and reaches ≈12.47%.
2–3	2–3	Correct formula but an incorrect weighting (e.g. 50/50, or debt twice equity), or the tax adjustment omitted.
0–1	0–1	No attempt, or the two costs simply averaged with no weighting.

Part (c) **10 Marks**

You are given the following information about Derek Ltd, a company financed entirely by equity capital.

	Derek Ltd
Number of equity shares of €1 ('000)	150,000
Market value per share, ex-dividend (€)	3.42
Current earnings (total) (€'000)	62,858
Current dividend (total) (€'000)	6,158
Statement of financial position value of capital employed (€'000)	315,000
Dividend five years ago (total) (€'000)	2,473

Estimate the cost of equity for Derek Ltd, using
(i) a historical growth rate and
(ii) the retention factor

FULL MODEL ANSWER

Setting up: dividend per share

D_0 per share = Total current dividend ÷ number of shares
= €6,158,000 ÷ 150,000,000 shares
D_0 ≈ €0.0411 per share

(i) Historical growth rate method

$g = (D_0 \div D_{-5})^{1/5} - 1$
$= (6{,}158 \div 2{,}473)^{0.2} - 1$
$= (2.4901)^{0.2} - 1$
g ≈ 20.02%

$K_e = D_1 \div P_0 + g = D_0(1+g) \div P_0 + g$
= (0.0411 × 1.2002) ÷ 3.42 + 0.2002
= 0.0144 + 0.2002
K_e ≈ 21.46%

This historical rate reflects Derek Ltd's *actual* dividend growth over the past 5 years, compounded — a genuinely high growth rate (20% p.a.) driven entirely by the specific dividend figures given. It assumes the past growth pattern continues, which is a real limitation if the last five years included an unusually strong (or weak) period not representative of the future.

(ii) Retention factor (br) method

Retention ratio b = (Earnings − Dividend) ÷ Earnings
= (62,858 − 6,158) ÷ 62,858 = **90.20%**
Return on capital employed r = Earnings ÷ Capital employed
= 62,858 ÷ 315,000 = **19.95%**
g = b × r = 90.20% × 19.95%
g ≈ 18.00%

$K_e = D_0(1+g) \div P_0 + g$
= (0.0411 × 1.1800) ÷ 3.42 + 0.1800
= 0.0142 + 0.1800
K_e ≈ 19.42%

The retention factor method gives a somewhat lower growth rate (18.00% vs. 20.02%) and correspondingly lower cost of equity (19.42% vs. 21.46%) than the historical method — this is a genuinely different growth estimate, not a rounding difference, since it's built on Derek Ltd's *current* retention and return-on-capital profile rather than its *past* dividend trajectory. The two methods won't generally agree exactly unless the company's payout ratio and return on capital have been perfectly stable over the whole 5-year window.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Correctly calculates D0 per share, correctly applies BOTH the historical growth formula (reaching ≈21.46% Ke) and the retention factor formula (reaching ≈19.42% Ke), and notes that the two methods needn't produce identical answers.
5–7	5–7	One method correctly applied, the other with an error (e.g. using earnings per share instead of dividend per share as D0, or the wrong base for r in the retention method).
2–4	2–4	Correct dividend per share calculated but errors in both growth rate methods.
0–1	0–1	No attempt, or no understanding of either growth estimation method.

Part (d) **10 Marks**

Calculate and comment upon some gearing ratios for CianPa plc. Extracts from CianPa plc Balance Sheet and Income Statement, 2022 are as follows:

	€m		€m
Non-current assets	96,804	Current liabilities of which	(24,000)
Current assets:		Short term borrowings	5,000
Inventory	288	Long term borrowings	(19,000)
Trade & other receivables	5,023	Other non-current liabilities	(5,580)
Taxation recoverable	21	Net assets	51,524
Cash and cash equivalents	7,481		
		EBIT	9,200
		Interest	(1,500)
		Market capitalisation	105,000

Note: Assume net assets equal shareholder funds.

FULL MODEL ANSWER

A source-data note before starting: summing the figures given, total assets (€96,804m + €12,813m current assets = €109,617m) less total liabilities (€24,000m + €19,000m + €5,580m = €48,580m) implies net assets of €61,037m — not the €51,524m stated. This looks like a transcription artefact in the source table rather than something with a clean resolution; since gearing ratios only need the borrowings, net assets, EBIT, interest and market capitalisation figures directly (not a fully reconciled balance sheet), the ratios below are calculated straight from the figures given for them, using the stated €51,524m net assets figure as instructed by the question's own note.

Total interest-bearing borrowings

Total borrowings = Short-term (€5,000m) + Long-term (€19,000m)
= €24,000m

Gearing Ratios

Ratio	Calculation	Result
Gearing (Debt ÷ Equity, book)	24,000 ÷ 51,524	46.58%
Gearing (Debt ÷ (Debt+Equity), book)	24,000 ÷ (24,000+51,524)	31.78%
Gearing (Debt ÷ Equity, market)	24,000 ÷ 105,000	22.86%
Gearing (Debt ÷ (Debt+Equity), market)	24,000 ÷ (24,000+105,000)	18.60%
Interest cover	9,200 ÷ 1,500	6.13 times

Comment

Book gearing is moderate; market gearing is considerably lower. CianPa's book-based gearing (46.58% debt-to-equity) looks like a reasonably geared company, but its market-based gearing (22.86%) is roughly half that. This gap arises because the market capitalisation (€105,000m) is more than double the book net assets (€51,524m) — the market is valuing CianPa well above its book value, which mechanically dilutes the apparent riskiness of its debt load when measured against that larger market equity base. This is worth flagging explicitly: book gearing significantly *overstates* the company's real financial risk from an investor's perspective, since market value, not book value, is what actually matters for assessing the risk borne by current shareholders.

Interest cover looks comfortable. At 6.13 times, EBIT comfortably covers the interest bill more than six times over, suggesting no immediate concern about CianPa's ability to service its debt from current operating earnings, even before considering its substantial cash balance (€7,481m) as an additional buffer.

Taken together, CianPa doesn't look excessively geared. Combining a healthy interest cover with a market gearing ratio well under 25% suggests the company likely has room to take on further debt if a good investment opportunity arose, without a material increase in financial distress risk — though this would need to be weighed against industry norms for whatever sector CianPa operates in.

How the 10 Marks Are Likely Allocated

Tier	Marks	What separates this tier
8–10	8–10	Calculates at least two gearing ratios correctly (ideally both book and market-based) plus interest cover, AND comments meaningfully on the gap between book and market gearing and what it implies about the company's real risk profile.
5–7	5–7	Correctly calculates the ratios but with limited or generic commentary, or calculates only book-based (or only market-based) gearing without both.
2–4	2–4	Attempts a gearing ratio but with a fundamental setup error (e.g. dividing by total assets rather than equity/debt+equity).
0–1	0–1	No attempt, or no genuine gearing ratio calculated.

Question 3 Total: 50 Marks